SAL College of Engineering

TUTORIAL

SEMESTER -4

BRANCH: CE/CSE/IT/ICT

Subject: Discrete Mathematics And Graph Theory

Subject Code: BE04000261

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DISCRETE MATHEMATICS AND GRAPH THEORY

TUTORIALS

TUTORIAL :1 SET THEORY

1 Define Power set and Give the power set of the following: {x,y,z}

2 Define Intersection of two sets, Union of two sets, Disjoint sets.

3 Define difference of two sets, Symmetric difference of two sets Show that for any two sets A and B, $A - (A \cap B) = A - B$.

4 Using venn diagram show that $A \cup (B' \cap C) = (A \cup B') \cap (AUC)$

5 Prove De-Morgan's Laws and $A \cap (B \cup C) \equiv (A \cap B) \cup (A \cap C)$

6 Given A = {x : x is an integer and $1 \leq x \leq 5$},
B = {3, 4, 5, 17}, and C = {1, 2, 3, . . .}, find $A \cap B$, $A \cup B$, $A \cap B \cap C$ and $A \cup C$.

7 Determine whether each of these statements is true or false.
1)$0 \in \emptyset$ 2)$\emptyset \subset \{0\}$ 3)$\{0\} \in \{0\}$
4) $\emptyset \in \{\emptyset\}$ 5)$\{\emptyset\} \in \{\{\emptyset\}\}$ 6) $\{\{\emptyset\}\} \subset \{\emptyset, \{\emptyset\}\}$

8 In a survey of 400 students in a school, 100 were found as drinking apple juice, 150 as drinking orange juice and 75 drinking both apple and orange juice. Find how many students drink neither apple nor orange juice.

9 For A = {2, 3, 4, 5, 6}, B = {3, 4, 5, 6, 7}, C = {4, 5, 6, 7, 8} find
a) $(A \cup B) \cap (A \cup C)$ b) $(A \cap B) \cup (A \cap C)$

10 How Many Integers between 1 to 2000 are divisible by 2,3,5 or 7

11 Among 100 students, 32 study mathematics, 20 study physics, 45 study biology, 15 study mathematics and biology, 7 study mathematics and physics, 10 study physics and biology and 30 do not study any of the three subjects.

a) Find the number of students studying all three subjects
b) Find the number of students studying exactly one of the three subjects

12 Consider a set of integers 1 to 500. Find

a) How many of these numbers are divisible by 3 or 5 or by 11
b) Also indicate how many are divisible by 3 or by 11 but not by all 3,5 and 11.
c) How many are divisible by 3 or 11 but not by 5.

13 Suppose that 100 mathematics students at a college taking at least one of the languages French, German, and Russian, given the following data:
65 study French, 45 study German, 42 study Russian, 20 study French and German, 25 study French and Russian, 15 study German and Russian.
(1) Find the number of students who study all the three languages.
(2) Find the number of students who study only French.

14 Prove that $A \cap \overline{B} = A \cap \overline{C}$ if and only if $A \cap B = A \cap C$.

15 In a class of 50 students, 12 enrolled for both Mathematics and Science, 32 enrolled for Science. If the students of the class enrolled for at least one of the two subjects, then how many students enrolled for only Mathematics but not Science?

TUTORIAL:2 FUNCTIONS

1 A function f: R+ → R is defined by f(x) = $x^2 - 8$. Check whether f is one-one and onto.

2 Show that the functions f(x) = x^3 and g(x) = $x^{1/3}$ for x ∈ R are inverses of one another.

3 Three functions f, g, h are defined from R → R as follows $f(x) = 2x^3 + 5$, $g(x) = cos\ x, h(x) = x^3 - 1$ find (1) $ho(gof)$ (2) $(hog)of$

4 Let f: R → R be function defined $f(X) = aX + b$. Find values of a and b for which $fof = I_R$

5 Define one-one function, onto function and inverse function.

6 Let function f: R → R be given by f(x) = $x^3 - 2$. Show that f^{-1} exists. Find f^{-1}.

7 Let X = {1,2,3}, f,g,h and s be function X to X given as
f = {(1,2),(2,3),(3,1)}
g= {(1,2),(2,1),(3,3)}
h={(1,1),(2,2),(3,1)}
s={(1,1),(2,2),(3,3)}
Find $fohog, gos, sos, fog, gof$ Are functions s and f invertible?

8 List all possible functions from $X = \{a, b, c\}$ to $Y = \{0,1\}$ and indicate in each case whether the function is one-to-one, is onto, and is one-to-one onto.

9 Let f: R → R be given by f(x) = 3X − 7. Determine whether function is Bijective or not. If yes then find its Inverse.

10 Determine weatherfunction f : N→N f(X) = $x^2 + 2$ is surjective or not.

11 Define the composition of two functions $f(x) = e^x$ and $g(x) = x^3$
Find $(f \circ g)(x)\ and\ (g \circ f)(x)$. Hence, show that $(f \circ g)(x) \neq (g \circ f)(x)$.

12 Show that the function f: R → $R^+ \cup \{0\}$ defined by $f(x) = |x|$ is not invertible. Modify the domain or codomain of f so that itbecomes invertible.

13 If $f: Q \to Q$ is defined as $f(x) = x^2$ then find $f^{-1}(16)$

14 If $f: R \to R$ and $g: R \to R$defined as $f(x) = cosx\ and\ g(x) = x^3$then find $(f \circ g)(x)\ and\ (g \circ f)(x)$

15 If $A = \{1,2,3,4\}\ and\ B = \{a, b, c, d\}$ and the function is defined as $f = \{(1, a), (2, b), (3, c), (4, c)\}$.Check whether the function is Many to One ,Into or not.

TUTORIAL:3 COUNTING

1 State the Pigeonhole Principle.

2 Compute the number of permutations of the set given {1,2,3,4,5}.

3 How many four digit numbers can be formed out of the digits 1,2,3,4,5,7,8,9,
   1) No repetition is permitted
   2) How many of these will be greater than 3000.

4 2 Mathematics papers and 5 other papers are to be arranged at an examination. Find the total number of ways if,
   1) Mathematics papers are consecutive
   2) Mathematics papers are not consecutive.

5 Determine the number of ways in which letters in the word PIONEER be arranged. So that two E's are always together.

6 How many diagonals does an n sided regular polygon have?

7 How many ways can one fill a box holding 100 pieces of candy from 30 different tyes of candy?

8 Show that $\binom{n}{1} + 6\binom{n}{2} + 6\binom{n}{3} = n^3$

9 A Group of 5 persons is to be formed from 6 men and 4 women. In how many way can this be done when

   (1) At least 2 women are included?
   (2) At most 2 women are included?

10 Find the number n of distinct permutations that can be formed from the letters of each word:(1)DISCRETE MATHEMATICS(2)SYMMETRY

11 In a box there are 5 black pens, 3 white pens and 4 red pens. In how many ways can 2 black pens, 2 white pens and 2 red pens can be chosen?

12 Find n if $\binom{n}{3} + \binom{n+2}{3} = {}^{n}P_{3}$

13 From 4 professors and 6 students, a committee of 3 is to be formed. In how many ways, this can be done, if the committee contains
   (1) at most 1 professor
   (2) at least 2 professors.

14 Among 100 people at least how many of them were born in the same month?

15 The Indian cricket team consist of 16 players. It includes 2 wicket keepers and 5 bowlers. In how many ways can cricket eleven be selected if we have select 1 wicket keeper and at least 4 bowlers?

TUTORIAL :4 PROPOSITIONAL LOGIC

1 Define Tautology and contradiction with examples. Prove that $P \rightarrow (P \vee Q)$ is tautology without constructing table.

2 Write a statement in English corresponding to the following symbolic statements:
P(x) : x is integer
Q(x) : x is a rational number
R(x) : x is prime integer
1) P(5)
2) $\forall x \sim P(x)$
3) $\exists x\ R(x)$
4) $\exists x \sim Q(x)$

3 Test the validity of the logical consequences: All dogs fetch.Ketty does not fetch. Therefore, Ketty is not a dog

4 Show the following equivalences without constructing the truth table
(i) $(\neg P \wedge (\neg Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R$
(ii) $\neg (P \wedge Q) \rightarrow (\neg P \vee (\neg P \vee Q)) \Leftrightarrow (\neg P \vee Q)$

5 Let I be the set of integers and R be relation "congruence modulo 5". Show that R is an equivalence relation. Determine the R-equivalence classes generated by the elements of I. Show that set of equivalence classes forms a partition of I

6 Find if the following is a tautology, contradiction or contingency.
$(p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$

7 Using truth table state whether each of the following implication is tautology.
a) $(p \wedge r) \rightarrow p$
b) $(p \wedge q) \rightarrow (p \rightarrow q)$
c) $p \rightarrow (p \vee q)$

8 Let P(x) be the statement "$x = x^2$ ". If the domain consists of the integers, what are the truth values of the following?
1) $\exists x\ P(x)$ 2) $\forall x\ \neg P(x)$ 3) $\exists x\ \neg P(x)$

9 Identify the error or errors in this argument that supposedly shows that if $\exists x\ P(x) \wedge \exists x\ Q(x)$ is true then $\exists x\ (P(x) \wedge Q(x))$ is true.
1. $\exists x\ P(x) \wedge \exists x\ Q(x)$ Premise
2. $\exists x\ P(x)$ Simplification from (1)
3. P(c) Existential instantiation from (2)
4. $\exists x\ Q(x)$ Simplification from (1)
5. Q(c) Existential instantiation from (2)
6. $P(c) \wedge Q(c)$ Conjunction from (3) and (5)
7. $\exists x\ (P(x) \wedge Q(x))$ Existential generalization

10 1) Use a truth table to verify the De Morgan's law $\neg(p \vee q) \equiv \neg p \wedge \neg q$
2) Show that $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$ is a tautology

TUTORIAL : 5 PREDICATE LOGICS

1 Using Predicate, Quantifier and rule of inference determine the given argument is valid or not “All student in the class understand logic. Xavier is a student in this class. Therefore, Xavier understand logic.”

2 Find conjunctive normal form and disjunctive normal form for the following without using truth table.
$(p \rightarrow q) \wedge (q \rightarrow p)$.

3 For an integer x prove that the following statements are equivalent: p: x is divisible by 10.q: x is divisible by 2 and 5.r: x is an even number and x is divisible by 5.

4 Rewrite the following statements using quantifier variables and predicate symbols:
1) All birds can fly.
2) Some women are genius.
3) There is a student who likes Discrete Mathematics but not Probability and Statistics.
4) Each integer is either even or odd.

5 Determine the validity of the argument given:
If I study, then I will not fail in Discrete Mathematics.
If I do not play cricket, then I will study.
But I failed in Discrete Mathematics.
Therefore I must have played cricket.

TUTORIAL :6 RELATIONS

1 Define Binary Relation, Reflexive Relation, Domain of a relation, Inverse of a relation and Complement of a relation.

2 Give example of relation which is both symmetric and anti-symmetric.

3 Show that if relation on a set is transistive and irreflexive then it is asymmetric.

4 Define the equivalence relation. Let $X = \{1,2,3,4\}$ $R = \{< x, y > \mid x > y\}$. Draw a graph of R and also give its matrix. Check whether the given relation an equivalence relation?

5 $A = \{1,2,3,5,6,10,15,30\} = B\ aRb\ if\ and\ only\ if\ f\ a|b$. Find the relation matrix.

6 $A = \{1,2,4,8\} aRb\ if and only\ if\ f\ a|b$ then find the digraph of relation.

7 Let R={(1,1),(2,2),(3,3),(1,2),(2,3),(3,1)}be a relation. Determine weather R is equivalence relation or not. Draw the difraph of this relation R.

8 Let R be a binary relation on the set of all positive integers such that,
$R = \{(a, b)|a - b\ is\ an\ odd\ positive\ integer\}$.
Is R reflexive ? symmetric ? Transistive? An equivalence relation?

9 Find the transistive closure of R by Warshall's algorithm. Where $A = \{1,2,3,4,5,6\}$ and $R = \{(x - y); |x - y| = 2\}$.

10 Let $S = \{1,2,3,4,5,6\}$. Determine whether or not each of the following is a partition of S.

(i) P1={(1,2,3),(1,4,5,6)}
(ii) P2={(1,2),(3,5,6)}
(iii) P3={(1,3,5).(2,4),(6)}
(iv) P4={(1,3,5),{2,4,6,7}}

11 Explain types of a Relation with a suitable example.

12 Find the transitive closure of
R = {(1,2), (3,4), (4,5), (4,1), (1,1)}. Where, A = {1,2,3,4,5}.

13 Let X = {1,2,3, ... ,7} and R = {(x, y): x − y is divisble by 3}. Show that R is an equivalence relation. Draw the graph of R.

14 Given S = {1, 2, 3, − − − − ,10} and a relation R on S. Where,
R = { ⟨x, y⟩ |x + y = 10} . What are the properties of relation R?

15 Let X = {1, 2, 3, 4, 5, 6, 7} and R = { ⟨x, y⟩ |x − y is divisible by 3}. Show that R is an equivalence relation on. Draw the graph of R.

TUTORIAL :7 PARTIAL ORDERING AND LATTICE

1 Define Posets and give example of poset.

2 Define Inverse order and dual of poset.

3 Draw the Hasse diagrams of the following sets under the partial ordering relation "divides" and indicate those which are totally ordered.

(i) {1,2,3,4} (ii) {3,5,15} (iii) {2,4,8,16} (iv) {1,2,3,6,12} (v) {2,3,6,12,24,36}

4 Define Lower bound and Upper bound. Let P = < 2, 3, 5, 6, 9, 15, 24, 45}, D> be a poset. Draw the Hasse diagram.
Find i) Maximal element. & minimal element.
ii) The greatest and least element.
iii) upper bound of { 9, 15 } ,and l.u.b of {9,15}
iv) lower bound of {15,24} and g.l.b of{15,24}

5 Draw Hasse-Diagram and also find Cover of each of the following: 1) (S_{70}, D) 2) (S_{105}, D) 3) (P(A), ⊆) Where A = {a,b,c}

6 When a poset is said to be a lattice? Explain. Is every poset a lattice? Justify. Is the poset {Φ, {p}, {q}, {p,q,r}, ⊆} a lattice?

7 Define the relation R on the set (Z) by aRb if a-b is non-negative even integer. Verify that R defines a partial order for Z. Is this partial order a total order?

8 Let A={a,b,c,d} and X be a relation on A. Whose matrix $M_R=\begin{bmatrix}1011\\0111\\0011\\0001\end{bmatrix}$

(i) Prove that R is partial order.
(ii) Draw the hass diagram of R.

9 Determine the matrix of the partial order whose Hass diagram is given in fig.1

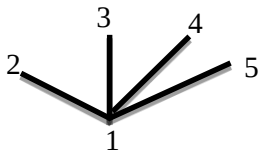


Fig.1

10 Define the greatest and lowest element . Define the upper bound and lower bound.

11 Let L be a lattice. Then a$\wedge b = a$ if and only if $a \vee b = b$.

12 Determine whether the following posets is boolean algebras. Justify your answer. A= {1,2,36} with divisibility.

13 Let A be a set of factors of positive integer m and relation is divisibility on A.For m = 45, show that POSET (A, $\leq$) is a Lattice.

14 Define: Bounded, Distributive and Complemented Lattices.

15 Draw the Hasse diagram of the set {1,3,9,18} under partial order relation

'divides' and indicate those which are chains.

TUTORIAL :8 RECURRENCE RELATION

1 Solve the recurrence relation : $a_{n+2} - 5a_{n+1} + 6a_n = 2$

2 Use generating function to solve a recurrence relation
$a_n = 3a_{n-1} + 2$ with $a_0 = 1$

3 Solve the recurrence relation $a_k - 5a_{k-1} + 6a_{k-2} = 0$, $a_0 = 2$, $a_1 = 5$

4 Solve the recurrence relation using the method of generating Function
$a_n - 5a_{n-1} + 6a_{n-2} = 3n \ with\ n \geq 2, and\ a_0 = 0, a_1 = 2$

5 Solve the recurrence relation $a_n = 2a_{n-1} + 3a_{n-2}, a_0 = 1, a_1 = 2$

6 Solve the recurrence relation
$a_n = 11a_{n-1} - 39a_{n-2} + 45a_{n-3}\ with\ a_0 = 5, a_1 = 11, a_2 = 25$

7 Solve the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2}, a_0 = 1, a_1 = 2$

8 Find the general Expression for a_n using Generating function to
$a_n - 7a_{n-1} + 12a_{n-2} = 0$ with $n \geq 2$

9 Draw a flow chart to calculate n! Using recursive procedure.

10 Define (1) Recursion (2) A primitive recursive function
Show that the function $f(x) = x + y$ is primitive recursive.

11 Define (1) A primitive recursive relation (2) A primitive recursive set
Show that$\{(x, x)/x \epsilon N\}$ which defines the relation of equality is primitive recursive relation.

12 What is Recursive Function? Write a Recursive algorithm to find out Fibonacci series.

13 Consider $a_r - 8a_{r-1} + 16a_{r-2} = 0$ where$a_2 = 16$ and $a_3 = 80$. Find the solution.

14 Find the general solution of $a_r - 3a_{r-1} - 4a_{r-2} = 4^r$.

15 Solve the following recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 2^n$ with initial conditions $a_0 = -1$ and $a_1 = 1$.

TUTORIAL :9 GROUP BASICS

1 Define ring, group, subgroup, Cyclic Group and abellian group.

2 Let (G, *) be a group. Let $= 5$. How many subgroups are there of G? Why?

3 Let $G = \{(a,b)|a,b \in R\}$. Define binary operation $(*)$ on G as $(a,b),(c,d) \in G,(a,b) * (c,d) = (ac, bc + d)$. Show that an algebraic structure (G,$*$) is a group.

4 Show that a group G is abelian iff $(ab)^2 = a^2b^2$ for all elements a and b in G.
Define Semi Group, Monoid. Give an example of an algebraic structure
Which is semi group but not monoid.

5 Let G be the set of non-zero real numbers. Define binary operation $(*)$ on G as $a * b = \frac{ab}{2}$. Show that an algebraic structure (G,$*$) is an abelian group.

TUTORIAL :10 NORMAL SUBGROUP AND LAGRANGE THEOREM, RINGS AND FIELD

1 Show that $< Z_6, +_6 >$ is a cyclic group of order 6 and also find its generators.

2 1) Show that every subgroup of a cyclic group is normal.
2) Define: Ring.
3) Write elements of the ring $\langle Z_{10}, +10, \times 10 \rangle$. And find $-3, -8, 3^{-1}, 4^{-1}$

3 Consider the set Q of a rational numbers. Let $*$ be the operation on Q defined by $a * b = a + b - ab$. Find (1) $3 * 4$ (2) the identity element for $*$.

4 Explain Cosets and Lagrange's theorem.

5 Find generators of $< Z_{10}, +_{10} >$ Also find its subgroups.

6 Define Left Coset and Right Coset. Let $H = \{[0][3]\}$ in Z_6 under addition. Find left cosets in $< Z_6, +_6 >$

7 Show that cyclic group is abelian but converse is not true. Is $< Z_5, +_5 >$ a cyclic group? If so, find its generators.

8 Show that (Z4, +4) is a cyclic group and find all the subgroups of $(Z_4, +4)$.

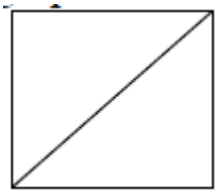
9 Show that if f: G→ $G'$is an isomorphism them f^{-1}: G' →G is also an isomorphism.

10 Prove that G = {1, 2, 3, 4, 5, 6} is a finite abelian group of order 6 with respect to multiplication modulo 7.

TUTORIAL : 11 BASICS OF GRAPH

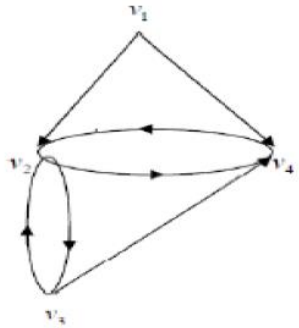
1 Define Isolated node, Pendent vertex, tree, Cycle and Length of a path with suitable example.

2 Define sub-graph. Find all sub-graphs for the given graph.

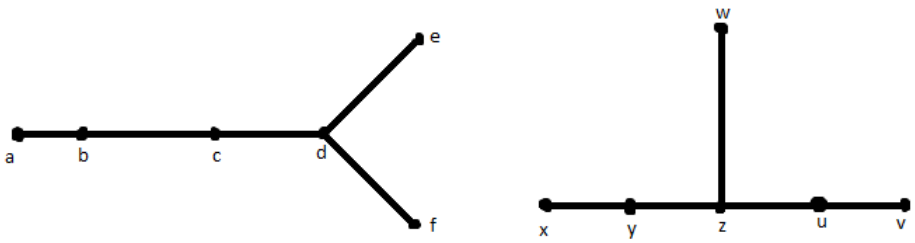


3 Draw diagraph and find in-degree and out-degree of each vertex from the given adjacency matrix. Use adjacency matrix, find total no. of path of length 1 and 2 between each vertex. $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

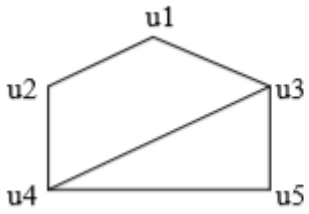
4 Find the elementary paths of lengths 1 and 2 from v1 to v4.

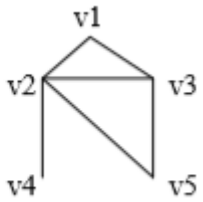


5 Determine whether the given pair of graph is Isomorphic or not?

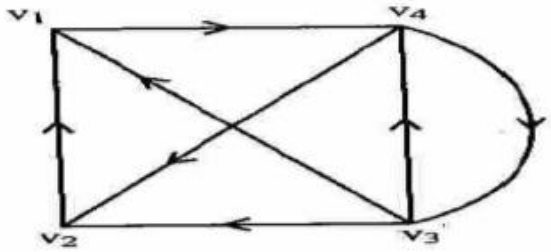


6 Define isomorphic graphs. State whether the following graphs are isomorphic or not.





7 Define adjacency matrix of a graph and obtain the adjacency matrix (A) for the following graph. State the in degree and out degree of all the vertices.



TUTORIAL : 12 GRAPH ALGORITHMS

1 Show that the maximum degree of any vertex in a simple graph with n vertices is (n-1) and for edges it is (n .(n-1))/2.

2 Determine the number of edged in a graph with 6 nodes, 2 of degree 4 and 4 of degree 2. Draw two such graphs.

3 It is possible to draw a simple graph with 4 vertices and 7 edges? Justify.

4 How many nodes are necessary to construct a graph with exactly 8 edges in which each node is of degree 2.

5 Draw a graph which is regular but not bipartite.

6 Define Eulerian graph, Planar Graph with proper illustrations.

7 Define a path in graph. Define length of the path. What is difference between a simple path and an elementary path?

TUTORIAL : 13 WEIGHTED GRAPHS

1 Explain the following terms with proper illustrations.
a) Directed graphs
b) Elementary path

2 Define the terms: Simple Graph, Multi – Graph, Weighted Graph, and Degree of a vertex, in degree and out degree of a vertex. Illustrate each with an example.

3 Define Cyclic graph, Null graph and Strongly connected graph with proper illustrations

4 Define adjacency matrix and path matrix of a graph with proper illustrations.

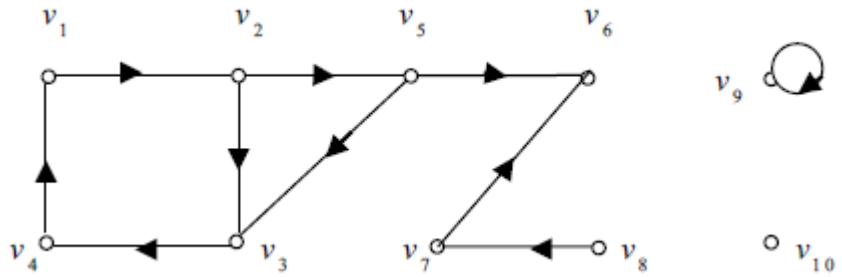
5 Explain Path and Circuit of a graph with proper illustrations.

6 Define regular graph and Complete graph. Is every regular graph is Complete?

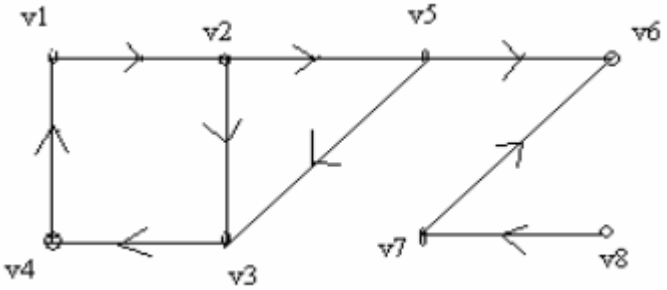
7 A Graph contains 21 edges and 3 vertices of degree 4 and all the other vertices of degree 2, find total number of vertices.

TUTORIAL :14 WARSHALL'S ALGORITHM AND REACHABILITY

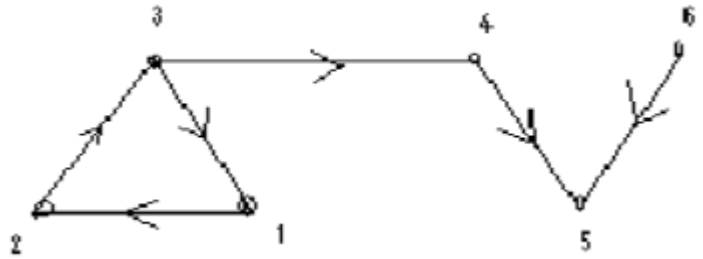
1 Find the reachable sets of (1) node v_1 and (2) node v_8 for the digraph given in following Figure.



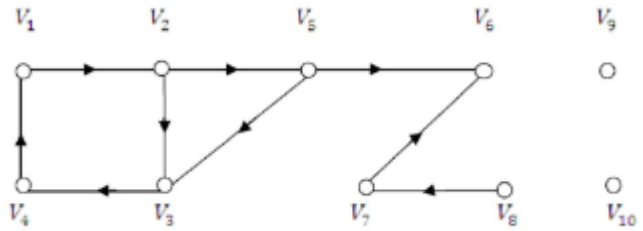
2 Define node base of a diagraph. Find all node base of the digraph given below



3 Define adjacency matrix of a graph and obtain the adjacency matrix (A) for the following graph. Find A^T. Also draw graph of A^T and find Path matrix P



4 Define node base of a simple digraph. Find reachablility set of all nodes for the followingdiagraph.



5 Find its transitive closure R by Warshall's algorithm. Where

A = {1, 2, 3, 4, 5}.
R= {(1,1),(1,2),(1,3),(1,4), (3,1),(3,2), (5,1), (5,2), (5,3), (5,4), (5,5)}

7 Use Warshall's algorithm to find the transitive closures of the relation
R = {(1,4), (2, 1), (2,3), (3,1), (3,4), (4,3)} on {1, 2, 3, 4}

TUTORIAL :15 TREES AND CONNECTEDNESS

1 Define Binary Tree , Tree anda directed tree

2 On a tree in the above question, perform the following traversals:
1. Pre-order traversal
2. Post-order traversal
3. In-order traversal

3 From the adjacency matrix of the simple digraph, how will you determine whether it is a directed tree? If it is a directed tree, how will you determine its root and terminal nodes?

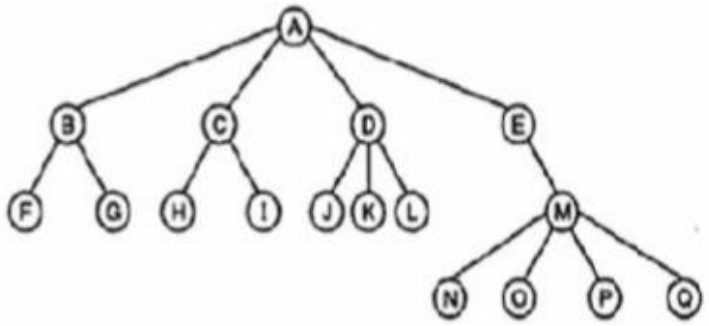
4 Describe link allocation techniques to represent binary trees.

5 Give other three representation of tree expressed by (v0(v1(v2)(v3)(v4))(v5(v6)(v7)(v8)(v9))(v10(v11)(v12))) obtain binary tree.

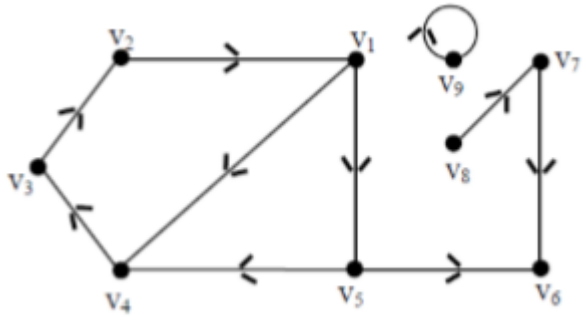
6 Define: Binary tree, Loop, Null graph. Show through two examples with n_t= 7 and n_t = 8 of complete binary trees that the total number of edges is given by $2(n_t - 1)$, where n_t is the number of terminal nodes.

7 Define Directed tree. Give three different tree representations of the following: (v0(v1(v2)(v3)(v4))(v5(v6)(v7)(v8)(v9))(v10(v11)(v12)))

8 Define Binary tree. Obtain Binary tree representation for the following:



9 Define node base of a digraph. Find all node base of the digraph in the figure given below. List out some of the properties of a node base.



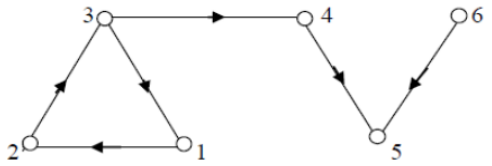
10 Define a directed tree. Show by means of an example that a simple digraph in which exactly one node has indegree 0 and every other node has indegree 1 is not necessarily a directed tree.

11 Define (i) A directed tree (ii) A Binary tree (iii) A complete -ary tree.

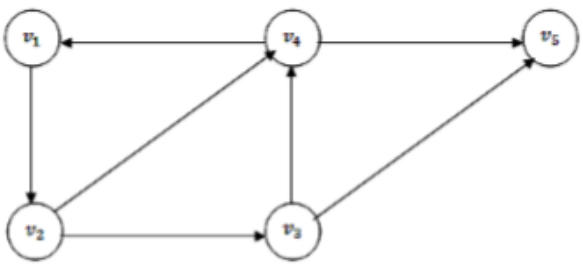
12 Show that in a complete binary tree the total number of edges is given by $2(n_i-1)$ Where is n_i the number of terminal nodes.

14 Define (i) An unilateral component. (ii) A strong component of a digraph.
Show that in a simple digraph G(V, E)
(i) every node of the digraph lies in exactly one strong component.
(ii) every node and every edge lies in at least one unilateral component.
(iii) every edge and every edge lies in exactly one weak component.

15 Define a unilateral component and strong component. Write unilateral and strong and weak components of the Graph given in following figure.



16 Find the strong components of the following diagraph. Also find its unilateral and weak components. Also find weather this graph is strongly connected, unilaterally connected, weakly connected.



17 Find all the indegrees and outdegrees of the nodes of the graph given in following figure. Give all the elementary cycles of this graph. List all the nodes which are reachable from another node of the diagraph

